Fake News Detection System

Kunj Bhesaniya
23b0995

Dhruvraj Merchant
23b1041

April 2025

Contents

1 Paper Description

- The paper proposes a fake news detection system using a hybrid deep learning model that combines Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory (BiLSTM) layers.
- It explores multiple word embedding techniques, including Word2Vec (CBOW and Skip-Gram) and GloVe, to effectively represent textual data.
- These embeddings are concatenated to capture rich semantic and contextual information from the news content.
- The model is evaluated on the WELFake dataset, comprising over 72,000 labeled news articles.
- Experimental results show that the proposed approach achieves high accuracy and outperforms traditional machine learning methods.

2 Model Architecture Overview

- **Input Layer:**
  - Preprocessed news text after cleaning (removal of noise, punctuation, URLs, etc.)
  - Lowercased and tokenized into word sequences
- **Word Embedding Layer:**
  - Four embedding strategies are explored:
    1. Word2Vec (CBOW) + Word2Vec (Skip-Gram)
    2. GloVe + Word2Vec (Skip-Gram)
    3. GloVe + Word2Vec (CBOW)
    4. GloVe only (Baseline)
  - Embeddings are concatenated to form dense input vectors
- **Convolutional Neural Network (CNN) Layer:**
  - Extracts fake news features from embedded sequences
- **Pooling Layer:**
  - Reduces feature map dimensions using max pooling
  - Preserves dominant features
- **Bidirectional LSTM (BiLSTM) Layer:**
  - Captures contextual dependencies in both forward and backward directions
- **Dropout Layer:**
  - Applied for regularization to reduce overfitting
- **Dense (Fully Connected) Layer:**
  - Fully connected layer activated by a sigmoid function
  - Outputs binary classification
- **Output Layer:**
  - Final prediction: 0 (Fake news), 1 (Real news)

3 Progress Done Till Now

3.1 Data Preprocessing and Cleaning

The dataset consists of news articles stored in the `title` and `text` columns. The following preprocessing steps were applied:

- Handled the nan values and converted all text to lowercase to ensure uniformity.
- Concatenated the title and text, which represents complete content of the news.
- Removed URLs, Hashtags, punctuations, alphanumeric tokens with digits, non printable characters and escape characters using regular expressions.
- tokenized the content into list of words for embedding it into a vector.

3.2 Word Embedding

- Each news article was converted into a vector representation using the Word2Vec model with the Continuous Bag of Words (CBOW) architecture.

Drive Link of the Work : Here

4 Further Plan of Action

- **Embedding Construction (April 15):**
  - Train or load embeddings: Word2Vec (Skip-Gram), GloVe
  - Build and save embedding matrices for all strategies
- **Model Architecture and Training (April 16-18 and May 1–2):**
  - Design and implement CNN + BiLSTM architecture
  - Prepare training pipeline (early stopping, checkpoints)
  - Train on Strategy 1 (CBOW + Skip-Gram)
  - Train on Strategy 2 (GloVe + Skip-Gram) and Strategy 3 (GloVe + CBOW)
- **Evaluation and Reporting (May 3):**
  - Train on Strategy 4 (GloVe only – baseline)
  - presentation report and demo preparation